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Lung Cancer Screening

Number: 0380

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Scope of Policy

This Clinical Policy Bulletin addresses lung cancer screening.

I. **Medical Necessity**

Aetna considers annual low-dose computed tomography (LDCT) scanning, also known as spiral CT or helical CT scanning, medically necessary for current or former smokers ages 50 to 80 years with a 20 pack-year or more smoking history and, if a former smoker, has quit within the past 15 years.

Aetna considers annual LDCT medically necessary for surveillance (starting 2 years after definitive treatment) of individuals with non-small cell lung cancer who have undergone definitive treatment.

II. **Experimental, Investigational, or Unproven**

The following lung cancer screening tests are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

- Artificial intelligence-based imaging for lung cancer screening;
- Computer-aided detection for chest radiographs for screening or diagnosis of lung cancer and for all other indications. There is presently inadequate evidence in the medical literature that population-based mass lung cancer screening with computer-aided detection for chest radiographs will contribute substantially to the detection of smaller cancers, or decreases mortality;
- LDCT as a screening test for all other indications not listed in Section I (e.g., asbestos-exposed individuals). **Note:** this does not apply to lung cancer surveillance;
- Positron emission tomography (PET) for lung cancer screening.

III. **Related Policies**

- CPB 0071 - Positron Emission Tomography (PET)

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Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered when selection criteria are met:	
71271	Computed tomography, thorax, low dose for lung cancer screening, without contrast material(s)
CPT codes not covered for indications listed in the CPB:	
***Artificial intelligence-based imaging*-no specific code**	
+ 0174T	Computer aided detection (CAD) (computer algorithm analysis of digital image data for lesion detection) with further physician review for interpretation and report, with or without digitization of film radiographic images, chest radiograph(s), performed concurrent with primary interpretation (Use 0174T in conjunction with 71010, 71020, 71021, 71022, 71030)
0175T	Computer aided detection (CAD) (computer algorithm analysis of digital image data for lesion detection) with further physician review for interpretation and report, with or without digitization of film radiographic images, chest radiograph(s), performed remote from primary interpretation (Do not report 0175T in conjunction with 71010, 71020, 71021, 71022, 71030)
71250	Computed tomography, thorax; without contrast material
78811 - 78813	Positron emission tomography (PET) imaging
78814 - 78816	Positron emission tomography (PET) imaging with concurrently acquired computed tomography (CT) for attenuation correction and anatomical localization imaging
Other CPT codes related to the CPB:	
71045 - 71048	Radiologic examination, chest
HCPCS codes covered if selection criteria are met:	
G0296	Counseling visit to discuss need for lung cancer screening using low dose CT scan (LDCT) (service is for eligibility determination and shared decision making)
ICD-10 codes covered if selection criteria are met:	
F17.210 - F17.219	Nicotine dependence, cigarettes
F17.290 - F17.299	Nicotine dependence, other tobacco product [pipes, cigars]
Z12.2	Encounter for screening for malignant neoplasm of respiratory organs
Z72.0	Tobacco use [former smokers ages 50-80 years]
Z87.891	Personal history of nicotine dependence [former smokers ages 50-80 years]
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	

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Code	Code Description
C34.10 – C34.92	Malignant neoplasm of bronchus and lung
F17.200 - F17.209, F17.220 - F17.229	Nicotine dependence, unspecified and chewing tobacco
Z77.090	Contact with and (suspected) exposure to asbestos [asbestos screening]

Surveillance of individuals with non-small cell cancer:

CPT codes covered when selection criteria are met:

71250	Computed tomography, thorax; without contrast material

ICD-10 codes covered if selection criteria are met:

C34.00 – C34.92	Malignant neoplasm of bronchus and lung [non-small cell lung cancer]
Z85.118	Personal history of other malignant neoplasm of bronchus and lung

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Background

Lung cancer is often diagnosed at a late stage; as a result, long term survival rates are poor. Detecting the disease and initiating treatment at an early stage are important for improving survival.

A clinical strategy for lung cancer detection that has demonstrated promise is low-dose computed tomography (LDCT), also known as spiral or helical CT scanning. LDCT is a radiographic technique that can provide high quality, three-dimensional images of the lungs during a single breath hold with less radiation exposure than conventional high resolution CT scanning. The purpose of LDCT screening is to identify the presence of lung cancer in an individual that does not demonstrate any symptoms.

Spiral Computed Tomography Scanning

Studies have shown that standard chest x-ray screening even when combined with sputum cytology does not decrease lung cancer mortality. Computed tomography (CT) is more sensitive in detecting parenchymal opacities than plain chest radiography; however, the expense, time, and radiation dose has prohibited CT from being considered of use as a screening modality. The latest generation of low-dose CT (LDCT) scanners (also known as spiral CT or helical CT) has the ability to scan the entire thorax in approximately 15 seconds, and the radiation dose used has been reduced to a level equivalent to mammography. Studies have demonstrated that spiral CT can detect small nodules in the lung that are otherwise poorly visible on chest X-ray.

The U.S. Preventive Services Task Force (USPSTF; Moyer, 2014) recommends annual screening for lung cancer with low-dose computed tomography (LDCT) in adults aged 55 to 80 years who have a 30 pack-year smoking history and currently smoke or have quit within the past 15 years. The Task Force stated that screening should be discontinued once a person has not smoked for 15 years or develops a health problem that substantially limits life expectancy or the ability or willingness to have curative lung surgery.

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The U.S. Preventive Services Task Force found adequate evidence that annual screening for lung cancer with LDCT in current and former smokers ages 55 to 80 years who have significant cumulative tobacco smoke exposure can prevent a substantial number of lung cancer deaths. THE USPSTF stated that direct evidence from the National Lung Screening Trial (NSLT, 2011), a large, well-conducted randomized, controlled trial (RCT) provides moderate certainty of the benefit of lung cancer screening with LDCT in this population.

A study published in the New England Journal of Medicine (NEJM) by the International Early Lung Cancer Action Program Investigators (2006) screened 31,567 asymptomatic persons at risk for lung cancer using low-dose CT from 1993 through 2005, and from 1994 through 2005, 27,456 repeated screenings were performed 7 to 18 months after the previous screening. These investigators estimated the 10-year lung-cancer-specific survival rate among participants with clinical stage I lung cancer that was detected on CT screening and diagnosed by biopsy, regardless of the type of treatment received, and among those who underwent surgical resection of clinical stage I cancer within 1 month. A pathology panel reviewed the surgical specimens obtained from participants who underwent resection. Screening resulted in a diagnosis of lung cancer in 484 participants. Of these participants, 412 (85 %) had clinical stage I lung cancer, and the estimated 10-year survival rate was 88 % in this subgroup (95 % confidence interval [CI]: 84 to 91). Among the 302 participants with clinical stage I cancer who underwent surgical resection within 1 month after diagnosis, the survival rate was 92 % (95 % CI: 88 to 95). The 8 participants with clinical stage I cancer who did not receive treatment died within 5 years after diagnosis. The authors concluded that annual spiral CT screening can detect lung cancer that is curable.

The editorial (Unger, 2006) that accompanied the NEJM study noted that "[a] troublesome problem in screening for lung cancer is the definition of a "high-risk" population – the population that could best benefit from lung cancer screening. The I-ELCAP study included more than 31,000 subjects who were at risk for lung cancer because they had a history of cigarette smoking or a history of occupational exposure (e.g., to asbestos, beryllium, uranium, or radon), or they had never smoked but had been exposed to second-hand smoke with or without a family history of lung cancer. The study was a systematic case-control observational study, not the gold-standard randomized trial .... The I-ELCAP study has considerable merit, but important questions remain. It is possible that without consideration of tumor biology, biases such as lead time and overdiagnosis could have been introduced in the final analysis of mortality. In the short run, chest CT scans alone do not reveal the differences between tumors and growing granulomatous lesions. Moreover, centrally located tumors or tumors located in the airway are not readily detectable by means of CT scanning. The question of cost-effectiveness remains unanswered."

Unger (2006) stated that "[w]e are making solid progress in combining CT scanning with sputum analysis, fluorescence bronchoscopy, and analysis of pulmonary fluids, exhaled gases, and blood by genomic, proteomic, and immunologic methods. Routine clinical applications of these methods, however, are not available. These technological wonders require extensive validation and proof that markers alone or in combination are sufficiently specific for the detection and diagnosis of lung cancer." An expert panel at the Radiological Society of North America's annual meeting (2006) did not endorse CT screening for lung cancer.

A recent study (Bach et al, 2007) reported that screening current or former smokers for lung cancer with CT increases the rate of diagnosis and treatment, but does not reduce the risk of advanced lung cancer or death from lung cancer. These findings imply that the additional small cancers detected by CT screening are unlikely to grow rapidly enough to significantly affect lung cancer mortality overall. These investigators analyzed data on 3,246 asymptomatic current or former smokers who were screened for lung cancer beginning in 1998. Participants received annual CT scans with comprehensive evaluation and treatment of detected nodules. Using a prediction model, these researchers examined the effect of CT screening on individuals by comparing the frequency of lung cancer detection, resection, advanced lung cancer cases, and deaths from lung cancer with what would have occurred in the absence of screening.

At a median follow-up of 3.9 years, there were 144 individuals diagnosed with lung cancer compared with 44.5 expected cases. There were 109 individuals who had a lung resection compared with 10.9 expected cases. There was no evidence of a decline in the number of diagnoses of advanced lung cancers (42 individuals compared with 33.4 expected cases) or deaths from lung cancer (38 deaths due to lung cancer

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observed and 38.8 expected). The authors stated that early detection and additional treatment did not save lives but did subject patients to invasive and possibly unnecessary treatments. The finding of a 10-fold increase in lung cancer surgeries resulting from screening underscores one of the potential public health consequences of CT screening. They noted that "if the majority of excess early cancers found through screening are unlikely to progress rapidly to a point where they cause clinically significant disease or death, then the thoracic surgeries performed to remove them may be insufficiently beneficial to justify the resulting morbidities. Until more conclusive data are available, asymptomatic individuals should not be screened outside of clinical research studies that have a reasonable likelihood of further clarifying the potential benefits and risks." This is in agreement with Black et al (2007) who stated that there is currently insufficient evidence that CT screening is clinically effective in reducing mortality from lung cancer.

These new findings are in contrast to the 2006 NEJM study, which concluded that CT screening could prevent 80 % of lung cancer deaths. The authors of that study had argued that a large RCT of CT screening be stopped, because the effectiveness of the method had already been proven. However, the authors of the current study disagree, stating, "we believe this method is not proven and should not be used broadly until a definitive randomized trial has been completed. That's in progress and will not be finished until 2009."

In an editorial that accompanied the study by Bach et al, Black and Baron (2007) stated that these findings present a stark contrast to those of the I-ELCAP study (International Early Lung Cancer Action Program Investigators, 2006) published 6 months earlier. The I-ELCAP investigators concluded from their findings that CT screening in populations at risk for lung cancer could prevent 80 % of lung cancer deaths. Black and Baron (2007) noted that because of the presence of a simulated control group, the measurement of mortality, and the completeness of the outcome assessment, the study by Bach et al more directly addresses the population effect of CT screening than does the ELCAP study.

An assessment by the California Technology Assessment Forum (CTAF, 2007) concluded that spiral CT for lung cancer screening did not meet CTAF's assessment criteria. The CTAF found: "None of the studies – even the most recent, large, international study – were designed to account for potential biases such as lead-time and length-time biases, and so cannot offer firm evidence that the ability of LDCT [low dose spiral computerized tomography] to detect small, early-stage cancers actually leads to decreased mortality. The one study which does compare mortality rates to a historical control did not find any survival advantage for those screened with LDCT. The risks of screening (radiation exposure, follow-up non-invasive and invasive procedures, anxiety) are potentially great, particularly if the benefits of screening are unproven. Thus, use of LDCT screening cannot yet be recommended outside of the investigational setting."

Infante et al (2008) stated that despite the high survival rates reported for screening-detected cases, the potential of screening of high-risk subjects for reducing lung cancer mortality is still unproven. These researchers herewith presented the baseline results of a randomized trial comparing screening for lung cancer with annual spiral computed tomography (CT) versus a yearly clinical review. Male subjects, 60 to 74 years old, and smokers of 20+ pack-years were enrolled. All subjects received a baseline medical examination, chest X-rays (CXR) and sputum cytology upon accrual. Participants randomized in the spiral CT group received a spiral CT scan at baseline, then yearly for the following 4 years. For controls, a yearly clinical examination was scheduled for the following 4 years. A total of 2,472 subjects were randomized (1,276 spiral CT arm, 1,196 controls). Age, smoking exposure and co-morbid conditions were similar in the 2 groups. In the spiral CT group, 28 lung cancers were detected, 13 of which were visible in the baseline chest X-rays (overall prevalence 2.2 %). A total of 16 out of 28 tumors (57 %) were stage I, and 19 (68 %) were resectable. In the control group, 8 cases were detected by the baseline chest X-rays (prevalence rate 0.67 %), 4 (50 %) were stage I, and 6 (75 %) were resectable. The authors concluded that baseline lung cancer detection rate in the spiral CT arm was higher than in most published studies. The stage I detection rate was increased 4-fold by spiral CT versus chest X-rays. However, more tumors in an advanced stage were also detected by CT. The high resection rate of screening-detected patients suggests a possible increase in cure rate. However, longer follow-up is needed for definitive conclusions. Furthermore, Smith and Berg (2008) stated that although screening with helical CT is currently under investigation in RCTs, observational studies have not shown evidence that it can detect lung cancer that is curable.

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Infante et al (2009) explored the effect of screening with low-dose spiral CT (LDCT) on lung cancer mortality. Secondary endpoints are incidence, stage at diagnosis, and resectability. Male subjects, aged 60 to 75 years, smokers of 20 or more pack-years, were randomized to screening with LDCT or control groups. All participants underwent a baseline, once-only chest X-ray and sputum cytology examination. Screening-arm subjects had LDCT upon accrual to be repeated every year for 4 years, whereas controls had a yearly medical examination only. A total of 2,811 subjects were randomized and 2,472 were enrolled (LDCT = 1,276; control = 1,196). After a median follow-up of 33 months, lung cancer was detected in 60 (4.7 %) patients receiving LDCT and 34 (2.8 %) control subjects (p = 0.016). Resectability rates were similar in both groups. More patients with stage I disease were detected by LDCT (54 % versus 34 %; p = 0.06) and fewer cases were detected in the screening arm due to intercurrent symptoms. However, the number of advanced lung cancer cases was the same as in the control arm. Twenty patients in the LDCT group (1.6 %) and 20 controls (1.7 %) died of lung cancer, whereas 26 and 25 died of other causes, respectively. The authors concluded that the mortality benefit from lung cancer screening by LDCT might be far smaller than anticipated.

Pastorino (2010) stated that lung cancer is the primary cause of cancer mortality in developed countries. First diagnosis only when disease has already reached the metastatic phase is the main reason for failure in treatment. In this regard, although low-dose spiral CT has proven to be effective in the early detection of lung cancer (providing both higher resectability and higher long-term survival rates), the capacity of annual CT screening to reduce lung cancer mortality in heavy smokers has yet to be demonstrated. Numerous ongoing large-scale RCTs are under way in high-risk individuals with different study designs. The initial results should be available within the next 2 years.

The National Lung Screening Trial Research Team (2011) noted that the National Lung Screening Trial (NLST) is a randomized multi-center study comparing low-dose helical CT with chest radiography in the screening of older current and former heavy smokers for early detection of lung cancer, which is the leading cause of cancer-related death in the United States. Five-year survival rates approach 70 % with surgical resection of stage IA disease; however, more than 75 % of individuals have incurable locally advanced or metastatic disease, the latter having a 5-year survival of less than 5 %. It is plausible that treatment should be more effective and the likelihood of death decreased if asymptomatic lung cancer is detected through screening early enough in its pre-clinical phase. For these reasons, there is intense interest and intuitive appeal in lung cancer screening with low-dose CT. The use of survival as the determinant of screening effectiveness is, however, confounded by the well-described biases of lead time, length, and over-diagnosis. Despite previous attempts, no test has been shown to reduce lung cancer mortality, an end point that circumvents screening biases and provides a definitive measure of benefit when assessed in a RCT that enables comparison of mortality rates between screened individuals and a control group that does not undergo the screening intervention of interest. The NLST is such a trial.

Jett and Midthun (2011) noted that screening for lung cancer is not currently recommended, even in persons at high-risk for this condition. Most patients with lung cancer present with symptomatic disease that is usually at an incurable, advanced stage. The recently reported NLST showed a 20 % decrease in deaths from lung cancer in high-risk persons undergoing screening with LDCT of the chest compared with chest radiography. The high-risk group included in the trial comprised asymptomatic persons aged 55 to 74 years, with smoking history of at least 30 pack-years. Screening with LDCT detected more cases of early-stage lung cancer and fewer cases of advanced-stage cancer, confirming that screening has shifted the stage of cancer at diagnosis and provides more persons with the opportunity for curative treatment. Although CT screening has risks and limitations, the 20 % decrease in deaths is the single most dramatic decrease ever reported for deaths from lung cancer, with the possible exception of smoking cessation. The authors stated that physicians should offer CT screening for lung cancer to patients who fit the high-risk profile defined in the NLST.

In contrast, Silvestri (2011) stated that after the publication of the NLST results, physicians will be faced with whether to begin ordering LDCT of the chest to screen for lung cancer in patients with a history of tobacco use. Despite the encouraging reduction in deaths observed by using LDCT in the NLST study population, recommending adoption of lung cancer screening in general practice is premature. Lessons learned from prostate and breast cancer screening should remind us that the reductions in deaths expected

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with screening are unfortunately not as readily achievable as initially believed. Furthermore, the potential harms of false-positive findings on chest CT are very real. The morbidity and even mortality associated with invasive diagnostic testing and surgical resection due to false- and true-positive findings on CT are likely to increase when the approach taken in the NLST is applied in non-specialty care settings and among the population at highest risk, namely, those with smoking-related co-morbid conditions. Although the NLST results are perhaps encouraging, they do not tell us enough that we can be sure that patients who undergo LDCT in an attempt to find early-stage lung cancer will have more benefit than harm.

In a position statement by the United Kingdom Lung Screen (UKLS) investigators following the NLST report, Field et al (2011) described the remaining questions that need to be answered by further research and to comment on the use of CT screening in the UK outside a clinical trial. The detailed design process of the UKLS protocol and international discussions were used to identify the research questions that remain to be answered and to inform those who may choose to consider offering CT screening, before these questions are answered. A series of research imperatives have been identified and these investigators advised that CT screening should be part of the ongoing clinical trial in the UK, currently in the pilot phase (UKLS). United Kingdom Lung Screen is randomizing 4,000 individuals for the pilot and a total of 32,000 for the main study. The authors concluded that there remain unresolved issues with respect to CT screening for lung cancer. These include its feasibility, psychosocial and cost-effectiveness in the UK, harmonization of CT acquisition techniques, management of suspicious screening findings, the choice of screening frequency and the selection of an appropriate risk group for the intervention.

In an editorial accompanying NSLT, Sox (2011) commented: "Policymakers should wait for cost-effectiveness analyses of the NLST data, further follow-up data to determine the amount of overdiagnosis in the NLST, and, perhaps, identification of biologic markers of cancers that do not progress. Modeling should provide estimates of the effect of longer periods of annual screening and the effect of better adherence to screening and diagnostic evaluation. Systematic reviews that include other, smaller lung-cancer screening trials will provide an overview of the entire body of evidence. Finally, it may be possible to define subgroups of smokers who are at higher or lower risk for lung cancer and tailor the screening strategy accordingly."

Saghir et al (2012) noted that the effects of LDCT screening on disease stage shift, mortality and over-diagnosis are unclear. These researchers reported lung cancer findings and mortality rates at the end of screening in the Danish Lung Cancer Screening Trial. A total of 4,104 men and women, healthy heavy smokers/former smokers were randomized to 5 annual LDCT screenings or no screening. Two experienced chest radiologists read all CT scans and registered the location, size and morphology of nodules. Nodules between 5 and 15 mm without benign characteristics were re-scanned after 3 months. Growing nodules (greater than 25 % volume increase and/or volume doubling time less than 400 days) and nodules greater than 15 mm were referred for diagnostic work-up. In the control group, lung cancers were diagnosed and treated outside the study by the usual clinical practice. Participation rates were high in both groups (screening: 95.5 %; control: 93.0 %; $p < 0.001$). Lung cancer detection rate was 0.83 % at baseline and mean annual detection rate was 0.67 % at incidence rounds ($p = 0.535$). More lung cancers were diagnosed in the screening group (69 versus 24, $p < 0.001$), and more were low-stage (48 versus 21 stage I-IIB non-small cell lung cancer (NSCLC) and limited stage small cell lung cancer (SCLC), $p = 0.002$), whereas frequencies of high-stage lung cancer were the same (21 versus 16 stage IIIA-IV NSCLC and extensive stage SCLC, $p = 0.509$). At the end of screening, 61 patients died in the screening group and 42 in the control group ($p = 0.059$); 15 and 11 died of lung cancer, respectively ($p = 0.428$). The authors concluded that CT screening for lung cancer brings forward early disease, and at this point no stage shift or reduction in mortality was observed. More lung cancers were diagnosed in the screening group, indicating some degree of over-diagnosis and need for longer follow-up.

Bach and colleagues (2012) conducted a systematic review of the evidence regarding the benefits and harms of lung cancer screening using LDCT. A multi-society collaborative initiative (involving the American Cancer Society, American College of Chest Physicians, American Society of Clinical Oncology, and National Comprehensive Cancer Network) was undertaken to create the foundation for development of an evidence-based clinical guideline. Medline (Ovid: January 1996 to April 2012), Embase (Ovid: January 1996 to April 2012), and the Cochrane Library (April 2012) were used for data selection. Of 591 citations identified and reviewed, 8 randomized trials and 13 cohort studies of LDCT screening met criteria for inclusion. Primary

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outcomes were lung cancer mortality and all-cause mortality, and secondary outcomes included nodule detection, invasive procedures, follow-up tests, and smoking cessation. Critical appraisal using pre-defined criteria was conducted on individual studies and the overall body of evidence. Differences in data extracted by reviewers were adjudicated by consensus. Three randomized studies provided evidence on the effect of LDCT screening on lung cancer mortality, of which the National Lung Screening Trial was the most informative, demonstrating that among 53,454 participants enrolled, screening resulted in significantly fewer lung cancer deaths (356 versus 443 deaths; lung cancer-specific mortality, 274 versus 309 events per 100,000 person-years for LDCT and control groups, respectively; relative risk, 0.80; 95 % CI: 0.73 to 0.93; absolute risk reduction, 0.33 %; p = 0.004). The other 2 smaller studies showed no such benefit. In terms of potential harms of LDCT screening, across all trials and cohorts, approximately 20 % of individuals in each round of screening had positive results requiring some degree of follow-up, while approximately 1 % had lung cancer. There was marked heterogeneity in this finding and in the frequency of follow-up investigations, biopsies, and percentage of surgical procedures performed in patients with benign lesions. Major complications in those with benign conditions were rare. The authors concluded that low-dose computed tomography screening may benefit individuals at an increased risk for lung cancer, but uncertainty exists about the potential harms of screening and the generalizability of results. The authors stated that "Screening a population of individuals at a substantially elevated risk of lung cancer most likely could be performed in a manner such that the benefits that accrue to a few individuals outweigh the harms that many will experience. However, there are substantial uncertainties regarding how to translate that conclusion into clinical practice".

Goulart et al (2012) noted that a recent randomized trial showed that LDCT screening reduces lung cancer mortality. Using data from the 2009 National Health Interview Survey, CMS, and the NLST, the authors performed an economic analysis of LDCT screening that includes a budget impact model, an estimate of additional costs per lung cancer death avoided attributed to screening, and a literature search of cost-effectiveness analyses of LDCT screening. They conducted a 1-way sensitivity analysis, reporting expenditures in 2011 U.S. dollars, and took the health care payer and patient perspectives. Low-dose CT screening will add $1.3 to $2.0 billion in annual national health care expenditures for screening uptake rates of 50 % to 75 %, respectively. However, LDCT screening will avoid up to 8,100 premature lung cancer deaths at a 75 % screening rate. The prevalence of smokers who qualify for screening, screening uptake rates, and cost of LDCT scan were the most influential parameters on health care expenditures. The additional cost of screening to avoid 1 lung cancer death is $240,000. Previous cost-effectiveness analyses have not conclusively shown that LDCT is cost-effective. The authors stated that LDCT screening may add substantially to the national health care expenditures. Although LDCT screening can avoid more than 8,000 lung cancer deaths per year, a cost-effectiveness analysis of the NLST will be critical to determine the value of this intervention and to guide decisions about its adoption.

Pinsky and Berg (2012) noted that the major NLST eligibility criteria were age 55 to 74 years, a 30 + pack year smoking history and current smoking status or having quit in the last 15 years. These investigators utilized data from SEER (Surveillance, Epidemiology and End Results), the U.S. Census and the National Health Interview Survey, as well as 2 statistical models of lung cancer risk, to estimate the proportion of the total U.S. population and of those currently diagnosed with lung cancer that would be covered by the NLST and other suggested eligibility criteria. For the NLST criteria, 26.7 % of lung cancers and 6.2 % of the population (over 40) were covered. A criterion of ever smokers aged 50 to 79 years would cover 68 % of the cancers while screening 30 % of the (over 40) population. To extend recommended screening beyond the NLST eligibility criteria, 2 questions are key. First, can the 20 % mortality reduction observed in NLST be extrapolated to populations at moderately lower risk? Second, given that such an extrapolation is valid, what background incidence rate is high enough for the balance between the benefits and harms of screening to be favorable? The authors stated that further research on these questions is needed.

In a case-control and prospective cohort study, Raji and associates (2012) evaluated the discrimination of the Liverpool Lung Project (LLP) risk model and demonstrated its predicted benefit for stratifying patients for CT screening by using data from 3 independent studies from Europe and North America. Participants in the European Early Lung Cancer (EUELC) and Harvard case-control studies and the LLP population-based prospective cohort (LLPC) study were included in this analysis. Main outcome measure was 5-year absolute risks for lung cancer predicted by the LLP model. The LLP risk model had good discrimination in both the Harvard (area under the receiver-operating characteristic curve [AUC], 0.76 [95 % CI: 0.75 to 0.78]) and the

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LLPC (AUC, 0.82 [CI, 0.80 to 0.85]) studies and modest discrimination in the EUELC (AUC, 0.67 [CI, 0.64 to 0.69]) study. The decision utility analysis, which incorporated the harms and benefit of using a risk model to make clinical decisions, indicated that the LLP risk model performed better than smoking duration or family history alone in stratifying high-risk patients for lung cancer CT screening. The authors concluded that validation of the LLP risk model in 3 independent external data sets demonstrated good discrimination and evidence of predicted benefits for stratifying patients for lung cancer CT screening. Moreover, they stated that further studies are needed to prospectively evaluate model performance and evaluate the optimal population risk thresholds for initiating lung cancer screening.

Pastorino et al (2012) stated that the efficacy and cost-effectiveness of LDCT screening in heavy smokers is currently under evaluation worldwide. These researchers' screening program started with a pilot study on 1,035 volunteers in Milan in 2000 and was followed-up in 2005 by a randomized trial comparing annual or biennial LDCT with observation, named Multicentric Italian Lung Detection (MILD), which included 4,099 participants, 1,723 randomized to the control group, 1,186 to biennial LDCT screening, and 1,190 to annual LDCT screening. Follow-up was stopped in November 2011, with 9,901 person-years for the pilot study and 17,621 person-years for MILD. A total of 49 lung cancers were detected by LDCT (20 in biennial and 29 in the annual arm), of which 17 were identified at baseline examination; 63 % were of stage I and 84 % were surgically resectable. Stage distribution and resection rates were similar in the 2 LDCT arms. The cumulative 5-year lung cancer incidence rate was 311/100,000 in the control group, 457 in the biennial, and 620 in the annual LDCT group ($p = 0.036$); lung cancer mortality rates were 109, 109, and 216/100,000 ($p = 0.21$), and total mortality rates were 310, 363, and 558/100,000, respectively ($p = 0.13$). Total mortality in the pilot study was similar to that observed in the annual LDCT arm at 5 years. The authors concluded that there was no evidence of a protective effect of annual or biennial LDCT screening. Furthermore, a meta-analysis of the 4 published randomized trials showed similar overall mortality in the LDCT arms compared with the control arm.

Ma and colleagues (2013) provided an estimate of the annual number of lung cancer deaths that can be averted by screening, assuming the screening regimens adopted in the NLST are fully implemented in the United States. The annual number of lung cancer deaths that can be averted by screening was estimated as a product of the screening effect, the U.S. population size (obtained from the 2010 US Census data), the prevalence of screening eligibility (estimated using the 2010 National Health Interview Survey [NHIS] data), and the lung cancer mortality rates among screening-eligible populations (estimated using the NHIS data from 2000 to 2004 and the third National Health and Nutrition Examination Survey linked mortality files). Analyses were performed separately by sex, age, and smoking status, with Poisson regression analysis used for mortality rate estimation. Uncertainty of the estimates of the number of avertable lung cancer deaths was quantified by simulation. Approximately 8.6 million Americans (95 % CI: 8.0 to 9.2 million), including 5.2 million men (95 % CI: 4.8 to 5.7 million) and 3.4 million women (95 % CI: 3.0 to 3.8 million), were eligible for lung cancer screening in 2010. If the screening regimen adopted in the NLST was fully implemented among these screening-eligible U.S. populations, a total of 12,250 (95 % CI: 10,170 to 15,671) lung cancer deaths (8,990 deaths in men and 3,260 deaths in women) would be averted each year. The authors concluded that the data from the current study indicate that LDCT screening could potentially avert approximately 12,000 lung cancer deaths per year in the U.S. Moreover, they stated that further studies are needed to estimate the number of avertable lung cancer deaths and the cost-effectiveness of LDCT screening under different scenarios of risk, various screening frequencies, and various screening uptake rates.

Aberle et al (2013) stated that the major harms of LDCT are radiation exposure, high false-positive rates, and the potential for over-diagnosis.

The American Cancer Society's guidelines on "Lung cancer screening" (Wender et al, 2013) provided the following recommendations:

Clinicians should ascertain the smoking status and smoking history of their patients aged 55 years to 74 years. Clinicians with access to high-volume, high-quality lung cancer screening and treatment centers should initiate a discussion about lung cancer screening with patients aged 55 years to 74 years who have at least a 30-pack-year smoking history, currently smoke, or have quit within the past 15 years, and who are in

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relatively good health. Core elements of this discussion should include the following benefits, uncertainties, and harms of screening:

- Benefit: Screening with low-dose computed tomography (LDCT) has been shown to substantially reduce the risk of dying from lung cancer.
- Limitations: LDCT will not detect all lung cancers or all lung cancers early, and not all patients who have a lung cancer detected by LDCT will avoid death from lung cancer.
- Harms: There is a significant chance of a false-positive result, which will require additional periodic testing and, in some instances, an invasive procedure to determine whether or not an abnormality is lung cancer or some non-lung cancer-related incidental finding. Fewer than 1 in 1,000 patients with a false-positive result experience a major complication resulting from a diagnostic work-up. Death within 60 days of a diagnostic evaluation has been documented, but is rare and most often occurs in patients with lung cancer.
- Chest x-rays (CXR) should not be used for cancer screening.

The American College of Chest Physicians' clinical practice guidelines on "Screening for lung cancer: Diagnosis and management of lung cancer" (Detterbeck et al, 2013) provided the following recommendations:

- In patients at risk for developing lung cancer, screening for lung cancer with CXR once or at regular intervals is not recommended (Grade 1A).
- In patients at risk for developing lung cancer, screening for lung cancer with sputum cytology at regular intervals is not suggested (Grade 2B).
- For smokers and former smokers who are age 55 to 74 and who have smoked for 30 pack-years or more and either continue to smoke or have quit within the past 15 years, annual screening with LDCT should be offered over both annual screening with CXR or no screening, but only in settings that can deliver the comprehensive care provided to National Lung Screening Trial (NLST) participants (Grade 2B). (Note: The most effective duration or frequency of screening is not known).
- For individuals who have accumulated fewer than 30 pack-years of smoking or are either younger than age 55 or older than 74, or individuals who quit smoking more than 15 years ago, and for individuals with severe co-morbidities that would preclude potentially curative treatment and/or limit life expectancy, computed tomography (CT) screening should not be performed (Grade 2C).

Yousaf-Khanand colleagues (2017) noted that in the USA annual lung cancer screening is recommended. However, the optimal screening strategy (e.g., screening interval, screening rounds) is unknown. These investigators provided results of the 4th screening round after a 2.5-year interval in the Dutch-Belgian Lung Cancer Screening trial (NELSON). Europe's largest, sufficiently powered randomized lung cancer screening trial was designed to determine whether LDCT screening reduces lung cancer mortality by greater than or equal to 25 % compared with no screening after 10 years of follow-up. The screening arm (n = 7,915) received screening at baseline, after 1 year, 2 years and 2.5 years. Performance of the NELSON screening strategy in the final fourth round was evaluated. Comparisons were made between lung cancers detected in the 1st 3 rounds, in the final round and during the 2.5-year interval. In round 4, a total of 46 cancers were screen-detected and there were 28 interval cancers between the 3rd and 4th screenings. Compared with the second round screening (1-year interval), in round 4 a higher proportion of stage IIIb/IV cancers (17.3 % versus 6.8 %, p = 0.02) and higher proportions of squamous-cell, broncho-alveolar and small-cell carcinomas (p = 0.001) were detected. Compared with a 2-year interval, the 2.5-year interval showed a higher non-significant stage distribution (stage IIIb/IV 17.3 % versus 5.2 %, p = 0.10). Additionally, more interval cancers manifested in the 2.5-year interval than in the intervals of previous rounds (28 versus 5 and 28 versus 19). The authors concluded that a 2.5-year interval reduced the effect of screening: the interval cancer rate was higher compared with the 1-year and 2-year intervals, and proportion of advanced disease stage in the final round was higher compared with the previous rounds.

Snowsill and colleagues (2018) examined the clinical effectiveness and cost-effectiveness of LDCT lung cancer screening in high-risk populations. Bibliographic sources included Medline, Embase, Web of Science and The Cochrane Library. These researchers carried out a systematic review of RCTs comparing LDCT screening programs with usual care (no screening) or other imaging screening programs [such as chest X-ray

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(CXR)]. Bibliographic sources included Medline, Embase, Web of Science and The Cochrane Library. Meta-analyses, including network meta-analyses, were performed. These investigators developed an independent economic model employing discrete event simulation and using a natural history model calibrated to results from a large RCT. There were 12 different population eligibility criteria and 4 intervention frequencies – single screen, triple screen, annual screening, and biennial screening; and a no-screening control arm. Clinical effectiveness – a total of 12 RCTs were included, 4 of which currently contributed evidence on mortality. Meta-analysis of these demonstrated that LDCT, with less than or equal to 9.80 years of follow-up, was associated with a non-statistically significant decrease in lung cancer mortality (pooled RR 0.94, 95 % CI: 0.74 to 1.19). The findings also showed that LDCT screening demonstrated a non-statistically significant increase in all-cause mortality. Given the considerable heterogeneity detected between studies for both outcomes, the results should be treated with caution. Network meta-analysis, including 6 RCTs, was performed to assess the relative clinical effectiveness of LDCT, CXR and usual care. The results showed that LDCT was ranked as the best screening strategy in terms of lung cancer mortality reduction; CXR had a 99.7 % probability of being the worst intervention and usual care was ranked second. Cost-effectiveness – screening programs were predicted to be more effective than no screening, reduce lung cancer mortality and result in more lung cancer diagnoses. Screening programs also increased costs. Screening for lung cancer was unlikely to be cost-effective at a threshold of £20,000/quality-adjusted life-year (QALY), but may be cost-effective at a threshold of £30,000/QALY. The incremental cost-effectiveness ratio for a single screen in smokers aged 60 to 75 years with at least a 3 % risk of lung cancer was £28,169 per QALY. Sensitivity and scenario analyses were conducted. Screening was only cost-effective at a threshold of £20,000/QALY in only a minority of analyses. The authors concluded that LDCT screening may be clinically effective in reducing lung cancer mortality, but there is considerable uncertainty. There is evidence that a single round of screening could be considered cost-effective at conventional thresholds, but there is significant uncertainty regarding the effect on costs and the magnitude of benefits.

The limitations of this study were as follows: Clinical effectiveness – the largest of the included RCTs compared LDCT with CXR screening rather than no screening; cost-effectiveness – a representative cost to the NHS of lung cancer has not been recently estimated according to key variables such as stage at diagnosis. Certain costs associated with running a screening program have not been included.

The USPSTF (2021) has revised the recommended ages and pack-years for lung cancer screening. It expanded the age range to 50 to 80 years (previously 55 to 80 years); and reduced the pack-year history to 20 pack-years of smoking (previously 30 pack-years).

Computer-Aided Detection for Chest Radiographs

Computer aided detection (CAD) systems are diagnostic tools that purportedly assist radiologists in the detection of subtle findings to facilitate early cancer detection. Used as an adjunct to radiographic or CT images of the chest, it analyzes and highlights areas in the image that appear to be solid nodules, alerting the radiologist to the need for additional analysis. The CAD system consists of dedicated computer software and a review workstation.

Computer-aided detection (CAD) has become one of the principal research areas in medical imaging and diagnostic radiology. It can be defined as diagnoses rendered by radiologists who utilize the output from computerized algorithm analyses of medical images as a second opinion in detecting lesions and in making diagnostic decisions. Presently, there are 2 diseases for which the United States Food and Drug Administration has given pre-market approval:

I. detection of breast cancer (adjunct to mammography), and
II. detection of signs consistent with lung cancer on chest radiographs.

I. detection of breast cancer (adjunct to mammography), and
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Current CAD schemes for the latter include nodule detection, interstitial disease detection, temporal subtraction, differential diagnosis of interstitial disease, and distinction between benign and malignant pulmonary nodules.

Available data on the use of CAD for detecting lung cancer appear to come mainly from one group of investigators (Abe, Doi, Kakeda, Shiraishi, and Suzuki). Their findings need to be further tested in clinical settings.

Coppini et al (2003) described a neural-network-based system for the CAD of lung nodules in chest radiograms. Images from the public Japanese Society of Radiological Technology (JSRT) database, including 247 radiograms, were used to build and test the system. These researchers performed a further test by using a second private database with 65 radiograms collected and annotated at the Radiology Department of the University of Florence. Both data sets included nodule and non-nodule radiographs. The use of a public data set along with independent testing with a different image set made the comparison with other systems easier and allowed a deeper understanding of system behavior. For the JSRT database, the authors observed that by varying sensitivity from 60 to 75 % the number of false alarms per image lies in the range 4 to 10, while accuracy is in the range 95.7 to 98.0 %. When the second data set was used, comparable results were obtained. These investigators concluded that observed system performances support the undertaking of system validation in clinical settings.

Sharsishi et al (2003) examined the effect of a high sensitivity in CAD for lung nodules in chest radiographs when extremely subtle cases were presented to radiologists. The chest radiographs used in this study consisted of 36 normal images and 54 abnormal images containing solitary lung nodules, of which 25 were extremely subtle and 29 were very subtle. Receiver operating characteristic (ROC) analysis for detecting lung nodules was performed with and without CAD. The levels of CAD output were simulated with a hypothetical ideal performance of 100 % sensitivity, but with 3 or 4 false positives per image. Six radiologists participated in an observer study in which cases were interpreted first without and then with the use of CAD. The average A(z) values for radiologists without and with CAD were 0.682 and 0.808, respectively. The performance of radiologists was improved significantly when high sensitivity was used ($p = 0.0003$). However, the radiologists were not able to recognize some extremely subtle nodules (5 of 54 nodules by all radiologists), even with the correct CAD output; these nodules were then considered as non-actionable. None of 306 computer-false positives was incorrectly regarded as a nodule by all radiologists, but 63 false positives were incorrectly identified by 1 or more radiologists. These investigators concluded that the accuracy of radiologists in the detection of some extremely subtle solitary pulmonary nodules can be improved significantly when the sensitivity of a CAD scheme can be made to be at an extremely high level. However, all of the 6 radiologists failed to identify some nodules (about 10 %), even with the correct output of the CAD.

Kakeda et al (2004) assessed the usefulness of a new commercially available CAD system with an automated method of detecting nodules due to lung cancers on chest radiograph. For patients with cancer, 45 cases with solitary lung nodules up to 25 mm in diameter (nodule size range, 8 to 25 mm in diameter; mean, 18 mm; median, 20 mm) were used. For healthy patients, 45 cases were selected on the basis of confirmation on chest CT. All chest radiographs were obtained with a computed radiography system. The CAD output images were produced with a newly developed CAD system, which consisted of an image server including CAD software called EpiSight/XR. Eight radiologists (4 board-certified radiologists and 4 radiology residents) participated in observer performance studies and interpreted both the original radiographs and CAD output images using a sequential testing method. The observers' performance was evaluated with ROC analysis. The average area under the curve value increased significantly from 0.924 without to 0.986 with CAD output images. Individually, the use of CAD output images was more beneficial to radiology residents than to board-certified radiologists. The authors concluded that this CAD system for digital chest radiographs can assist radiologists and has the potential to improve the detection of lung nodules due to lung cancer.

Suzuki et al (2005) developed a technique that uses a multiple massive-training artificial neural network (multi-MTANN) to reduce the number of false-positive results in a CAD scheme for detecting nodules in chest radiographs. These investigators found that use of the trained multi-MTANN eliminated 68.3 % of

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false-positive findings with a reduction of 1 true-positive result. The false-positive rate of the original CAD scheme was improved from 4.5 to 1.4 false positives per image, at an overall sensitivity of 81.3 %, suggesting that this technique reduced the false-positive rate of the CAD scheme for lung nodule detection on chest radiographs, while maintaining a level of sensitivity.

Doi (2005) stated that because CAD can be applied to all imaging modalities, all body parts, and all kinds of examinations, it is likely that CAD will have a major impact on medical imaging and diagnostic radiology in the 21st century.

Li et al (2008) retrospectively examined the sensitivity of and number of false-positive marks made by a commercially available CAD system for identifying lung cancers previously missed on chest radiographs by radiologists, with histopathological results as the reference standard. A CAD nodule detection program was applied to 34 postero-anterior digital chest radiographs obtained in 34 patients (13 women, 21 men; mean age of 69 years). All 34 radiographs showed a nodular lung cancer that was apparent in retrospect but had not been mentioned in the report. Two radiologists identified these radiologist-missed cancers on the chest radiographs and graded them for visibility, location, subtlety (extremely subtle to extremely obvious on a 10-point scale), and actionability (actionable or not actionable according to whether the radiologists probably would have recommended follow-up if the nodule had been detected). The CAD results were analyzed to determine the numbers of cancers and false-positive nodules marked and to correlate the CAD results with the nodule grades for subtlety and actionability. The chi-2 test or Fisher exact test for independence was used to compare CAD sensitivity between the very subtle (grade 1 to 3) and relatively obvious (grade greater than 3) cancers and between the actionable and not actionable cancers. The CAD program had an overall sensitivity of 35 % (12 of 34 cancers), identifying 7 (30 %) of 23 very subtle and 5 (45 %) of 11 relatively obvious radiologist-missed cancers ($p = 0.21$) and detecting 2 (25 %) of 8 missed not actionable and 10 (38 %) of 26 missed actionable cancers ($p = 0.33$). The CAD program made an average of 5.9 false-positive marks per radiograph.

White and associates (2009) examined the ability of a CAD system to detect lung cancer overlooked at initial interpretation by the radiologist. In patients with lung cancer diagnosed from 1995 to 2006 at 2 institutions, each chest radiograph obtained prior to tumor discovery was evaluated by 2 radiologists for an overlooked lesion. The size and location of the nodules were documented and graded for subtlety (grades 1 to 4, 1 = very subtle). Each radiograph with a missed lesion was analyzed by a commercial CAD system, as was the follow-up image at diagnosis. An age-matched and sex-matched control group was used to assess CAD false-positive rates. Missed lung cancer was found in 89 patients (age range of 51 to 86 years; mean age of 65 years; 9 women, 80 men) on 114 radiographs. Lesion size ranged from 0.4 to 5.5 cm (mean of 1.8 cm). Lesions were most commonly peripheral ($n = 63$, 71 %) and in upper lobes ($n = 67$, 75 %). Lesion subtlety score was 1, 2, 3, or 4 on 43, 49, 17, and 5 radiographs, respectively. Computer-aided detection identified 53 (47 %) and 46 (52 %) undetected lesions on a per-image and per-patient basis, respectively. The average size of lesions detected with CAD was 1.73 cm compared with 1.85 cm for lesions that were undetected ($p = 0.47$). A significant difference ($p = 0.017$) was found in the average subtlety score between detected lesions (score = 2.06) and undetected lesions (score = 1.68). An average of 3.9 false-positive results occurred per radiograph; an average of 2.4 false-positive results occurred per radiograph for the control group. The authors concluded that CAD has the potential to detect approximately 50 % of the lesions overlooked by human readers at chest radiography.

Yanagawa and co-workers (2009) assessed the performance of a commercially available CAD system in the detection of pulmonary nodules with or without ground-glass opacity (GGO) using 64-detector-row CT compared to visual interpretation. Computed tomographic examinations were performed on 48 patients with existing or suspicious pulmonary nodules on chest radiography. Three radiologists independently reported the location and pattern (GGO, solid, or part solid) of each nodule candidate on CT scans, assigned each a confidence score, and then analyzed all scans using the CAD system. A reference standard was established by a consensus panel of different radiologists, who found 229 non-calcified nodules with diameters greater than or equal to 4 mm. True-positive and false-positive results and confidence levels were used to generate jackknife alternative free-response receiver-operating characteristic plots. The sensitivity of GGO for 3 radiologists (60 % to 80 %) was significantly higher than that for the CAD system (21%) (McNemar's test, $p < 0.0001$). For overall and solid nodules, the figure-of-merit values without and with the CAD system were

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significantly different (p = 0.005 to 0.04) on jackknife alternative free-response receiver-operating characteristic analysis. For GGO and part-solid nodules, the figure-of-merit values with the CAD system were greater than those without the CAD system, indicating no significant differences. The authors concluded that radiologists are significantly superior to this CAD system in the detection of GGO, but the CAD system can still play a complementary role in detecting nodules with or without GGO.

De Boo and colleagues (2009) stated that detection of focal pulmonary lesions is limited by quantum and anatomical noise and highly influenced by variable perception capacity of the reader. Multiple studies have proven that lesions – missed at time of primary interpretation – were visible on the chest radiographs in retrospect. Computer-aided diagnosis schemes do not alter the anatomical noise but aim at decreasing the intrinsic limitations and variations of human perception by alerting the reader to suspicious areas in a chest radiograph when used as a "second reader". Multiple studies have shown that the detection performance can be improved using CAD especially for less experienced readers at a variable amount of decreased specificity. There seem to be a substantial learning process for both, experienced and inexperienced readers, to be able to optimally differentiate between false-positive and true-positive lesions and to build up sufficient trust in the capabilities of these systems to be able to use them at their full advantage. Studies so far focused on stand-alone performance of the CAD schemes to reveal the magnitude of potential impact or on retrospective evaluation of CAD as a second reader for selected study groups. The authors stated that more research is needed to evaluate the performance of these systems in clinical routine and to examine the trade-off between performance increase in terms of increased sensitivity and decreased inter-reader variability and loss of specificity and secondary indicated follow-up examinations for further diagnostic work-up.

Way and colleagues (2010) assessed the effect of CAD on radiologists' estimates of the likelihood of malignancy of lung nodules on CT imaging. A total of 256 lung nodules (124 malignant and 132 benign) were retrospectively collected from the thoracic CT scans of 152 patients. An automated CAD system was developed to characterize and provide malignancy ratings for lung nodules on CT volumetric images. An observer study was conducted using ROC analysis to evaluate the effect of CAD on radiologists' characterization of lung nodules. Six fellowship-trained thoracic radiologists served as readers. The readers rated the likelihood of malignancy on a scale of 0 % to 100 % and recommended appropriate action first without CAD and then with CAD. The observer ratings were analyzed using the Dorfman-Berbaum-Metz multi-reader, multi-case method. The CAD system achieved a test area under the ROC curve (A(z)) of 0.857 +/- 0.023 using the perimeter, 2 nodule radii measures, 2 texture features, and 2 gradient field features. All 6 radiologists obtained improved performance with CAD. The average A(z) of the radiologists improved significantly ($p < 0.01$) from 0.833 (range of 0.817 to 0.847) to 0.853 (range of 0.834 to 0.887). The authors concluded that CAD has the potential to increase radiologists' accuracy in assessing the likelihood of malignancy of lung nodules on CT imaging.

de Hoop et al (2010) evaluated how CAD affects reader performance in detecting early lung cancer on chest radiographs. In this ethics committee-approved study, 46 individuals with 49 CT-detected and histologically proved lung cancers and 65 patients without nodules at CT were retrospectively included. All subjects participated in a lung cancer screening trial. Chest radiographs were obtained within 2 months following screening CT. Four radiology residents and 2 experienced radiologists were asked to identify and localize potential cancers on the chest radiographs, first without and subsequently with the use of CAD software. A figure of merit was calculated by using free-response ROC analysis. Tumor diameter ranged from 5.1 to 50.7 mm (median of 11.8 mm). Fifty-one % (22 of 49) of lesions were subtle and detected by 2 or fewer readers. Stand-alone CAD sensitivity was 61 %, with an average of 2.4 false-positive annotations per chest radiograph. Average sensitivity was 63 % for radiologists at 0.23 false-positive annotations per chest radiograph and 49 % for residents at 0.45 false-positive annotations per chest radiograph. Figure of merit did not change significantly for any of the observers after using CAD. Computer-aided detection marked between 5 and 16 cancers that were initially missed by the readers. These correctly CAD-depicted lesions were rejected by radiologists in 92 % of cases and by residents in 77 % of cases. The authors concluded that the sensitivity of CAD in identifying lung cancers depicted with CT screening was similar to that of experienced radiologists. However, CAD did not improve cancer detection because, especially for subtle lesions, observers were unable to sufficiently differentiate true-positive from false-positive annotations.

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The American College of Radiology's Appropriateness Criteria® screening for pulmonary metastases (Mohammed et al, 2010) stated that "[c]omputer-aided detection (CAD) for pulmonary metastatic disease has been adapted to chest CT from applications from mammography. Although these programs are in their developmental phases, it has been suggested that CAD can be used as a second look after the radiologist has completed reviewing the study. Nevertheless, these programs require more development and currently can only be used when there is limited breathing artifact and stable lung expansion. A recent study demonstrated that CAD detected 82.4 % of known pulmonary nodules under ideal conditions. CAD is still in the experimental phase and currently has limited use in evaluating patients with pulmonary metastatic disease".

Mazzone et al (2013) stated that the sensitivity of CT-based lung cancer screening for the detection of early lung cancer is balanced by the high number of benign lung nodules identified, the unknown consequences of radiation from the test, and the potential costs of a CT-based screening program. Computer-aided detection chest radiography may improve the sensitivity of standard chest radiography while minimizing the risks of CT-based screening. Study subjects were age 40 to 75 years with 10+ pack-years of smoking and/or an additional risk for developing lung cancer. Subjects were randomized to receive a PA view chest radiograph or placebo control (went through the process of being imaged but were not imaged). Images were reviewed first without then with the assistance of CAD. Actionable nodules were reported and additional evaluation was tracked. The primary outcome was the rate of developing symptomatic advanced stage lung cancer. A total of 1,424 subjects were enrolled; 710 received a CAD chest radiograph, 29 of whom were found to have an actionable lung nodule on prevalence screening. Of the 15 subjects who had a chest CT performed for additional evaluation, a lung nodule was confirmed in 4, 2 of which represented lung cancer. Both of the cancers were seen by the radiologist unaided and were identified by the CAD chest radiograph. The cumulative incidence of symptomatic advanced lung cancer was 0.42 cases per 100 person-years in the control arm; there were no events in the screening arm. The authors concluded that further evaluation is needed to determine if CAD chest radiography has a role as a lung cancer screening tool.

Massion et al (2020) noted that the management of indeterminate pulmonary nodules (IPNs) remains challenging, resulting in invasive procedures and delays in diagnosis and treatment. Strategies to lower the rate of unnecessary invasive procedures and optimize surveillance regimens are needed. These researchers developed and validated a deep learning (DL) method to improve the management of IPNs. A Lung Cancer Prediction Convolutional Neural Network model was trained using computed tomography (CT) images of IPNs from the National Lung Screening Trial, internally validated, and externally tested on cohorts from 2 academic centers. The areas under the receiver operating characteristic curve (AUC) in the external validation cohorts were 83.5 % (95 % confidence interval [CI]: 75.4 % to 90.7 %) and 91.9 % (95 % CI: 88.7 % to 94.7 %), compared with 78.1 % (95 % CI: 68.7 % to 86.4 %) and 81.9 % (95 % CI: 76.1 % to 87.1 %), respectively, for a commonly used clinical risk model for incidental nodules. Using 5 % and 65 % malignancy thresholds defining low- and high-risk categories, the overall net re-classifications in the validation cohorts for cancers and benign nodules compared with the Mayo model were 0.34 (Vanderbilt) and 0.30 (Oxford) as a rule-in test, and 0.33 (Vanderbilt) and 0.58 (Oxford) as a rule-out test. Compared with traditional risk prediction models, the Lung Cancer Prediction Convolutional Neural Network was associated with improved accuracy in predicting the likelihood of disease at each threshold of management and in the external validation cohorts. The authors concluded that this study showed that this DL algorithm could correctly re-classify IPNs into low- or high-risk categories in more than 1/3 of cancers and benign nodules when compared with conventional risk models, potentially reducing the number of unnecessary invasive procedures and delays in diagnosis. Moreover, these researchers stated that their model is intended to be improved over time as data collections are added and structured curation efforts continue. They noted that although more stringent clinical validations on additional (external and independent) datasets are needed, these findings suggested that it may be possible to address a major problem in the management of individuals presenting with IPNs by using an ML-derived prediction model.

The authors stated that this study had several drawbacks. First, although these investigators compared the performance of LCP-CNN with that of relevant clinical risk models, they did not report its potential to change clinical decision-making. Because some clinical parameters were missing, not all risk models could be run on all datasets. In the future, comparisons with multiple models would be desirable. Second, because of the smaller size of the Vanderbilt University Medical Center (VUMC) dataset (n = 116), the difference in AUC was not significant (p = 0.082), although all VUMC re-classification results were significant. Third,

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despite the differences in disease prevalence and patient populations across the 3 validation datasets, the same linear calibration between the LCP-CNN and risk was used for all the results; however, the results may be further optimized by a population specific calibration. For example, although the re-classification of VUMC and Oxford University Hospitals National Health Service Foundation Trust (OUH) datasets was very good, on the U.S. National Lung Screening Trial (NLST), 3.5 % of controls were incorrectly classified as intermediate risk compared with Brock, because of the low prevalence of disease. The OUH dataset did not capture the patients' history of cancer, which is necessary to calculate the Mayo risk scores, although patients who had received a cancer diagnosis in the last 5 years were excluded. Thus, in calculating the Mayo scores, it was assumed that the OUH patients had no history of cancer. Fourth, although the results were at the nodule level rather than the patient level, the VUMC dataset only had 1 nodule per patient, and the mean number of nodules per patient in the OUH dataset was 1.08.

Kim et al (2022) stated that limited data are available regarding whether computer-aided diagnosis (CAD) would improve assessment of malignancy risk in PNs. In a retrospective, multi-reader, multi-case study, these investigators examined the effect of an artificial intelligence (AI)-based CAD tool on clinician IPN diagnostic performance and agreement for both malignancy risk categories and management recommendations. This trial was carried out in June and July 2020 on chest CT studies of IPNs. Readers used only CT imaging data and provided an estimate of malignancy risk and a management recommendation for each case without and with CAD. The effect of CAD on average reader diagnostic performance was assessed using the Obuchowski-Rockette and Dorfman-Berbaum-Metz method to calculate estimates of AUC, sensitivity, and specificity. Multi-rater Fleiss κ statistics were used to measure inter-observer agreement for malignancy risk and management recommendations. A total of 300 chest CT scans of IPNs with maximal diameters of 5 mm to 30 mm (50.0 % malignant) were reviewed by 12 readers (6 radiologists, and 6 pulmonologists) (patient median age of 65 years; inter-quartile range [IQR], 59 to 71 years; 164 [55 %] men). Readers' average AUC improved from 0.82 to 0.89 with CAD ($p < 0.001$). At malignancy risk thresholds of 5 % and 65 %, use of CAD improved average sensitivity from 94.1 % to 97.9 % ($p = 0.01$) and from 52.6 % to 63.1 % ($p < 0.001$), respectively. Average reader specificity improved from 37.4 % to 42.3 % ($p = 0.03$) and from 87.3 % to 89.9 % ($p = 0.05$), respectively. Reader inter-observer agreement improved with CAD for both the less than 5 % (Fleiss κ, 0.50 versus 0.71; $p < 0.001$) and more than 65 % (Fleiss κ, 0.54 versus 0.71; $p < 0.001$) malignancy risk categories. Overall reader inter-observer agreement for management recommendation categories (no action, CT surveillance, diagnostic procedure) also improved with CAD (Fleiss κ, 0.44 versus 0.52; $p = 0.001$). The authors concluded that the use of CAD improved estimation of IPN malignancy risk on chest CT scans and improved inter-observer agreement for both risk stratification and management recommendations. Moreover, these researchers stated that these findings provided crucial support for bringing CAD tools closer to clinical implementation for IPN risk stratification. In addition, these findings suggested that the LCP-CNN CAD tool may have a meaningful impact on subsequent management decisions. These investigators stated that future prospective studies are needed to examine the effect of CAD on clinical and patient-centered outcomes in real-world settings.

The authors stated that this study had several drawbacks. First, readers were not provided any clinical information when assessing IPN imaging data; therefore, generalizability to a routine clinical setting was limited. The authors' intention was to avoid introducing bias and to exclude the uncertainty of whether variability in image interpretation was because of clinical context rather than nodule characteristics. The LCP-CNN CAD tool estimated malignancy risk based on imaging features without consideration of other clinical information (e.g., age, smoking history); thus, their objective was to determine its impact on clinicians' ability to evaluate IPNs in the absence of other risk factors. Second, before reviewing any cases, readers were told that the prevalence of malignancy was higher than is normally found in clinical practice, potentially introducing context bias and inflating all risk estimates. Third, the modest number of part-solid nodules included in this study limited the generalizability of these findings to this subgroup of pulmonary nodules. Fourth, the LCP scores provided to readers were not accompanied by measures of uncertainty. Future studies should further examine the reliability of the LCP-CNN CAD tool. Fifth, although these researchers observed significant improvements in diagnostic performance for each reader, the absolute increases in AUC across readers varied, and further investigation is needed to quantify what constitutes a clinically important improvement in discrimination. Sixth, as with all CAD-based studies, these findings were applicable only to this software system, and other systems should not be assumed to produce similar results.

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In summary, while CAD for chest radiographs may be potentially useful in screening lung cancer, its clinical value needs to be established by RCTs.

Positron Emission Tomography (PET)

Chien et al (2013) stated that although LDCT is a recommended modality for lung cancer screening in high-risk populations, the role of other modalities, such as [(18)F]fluorodeoxyglucose-positron emission tomography (PET), is unclear. These investigators conducted a systematic review to describe the role of PET in lung cancer screening. A systematic review was conducted by reviewing primary studies focusing on PET screening for lung cancer until July 2012. Two independent reviewers identified studies that were compatible for inclusion/exclusion criteria. The analysis was restricted to English and included studies published since 2000. A descriptive analysis was used to summarize the results, and the pooled diagnostic performance of selective PET screening was calculated by weighted average using individual sample sizes. Among the identified studies (n = 3,497), 12 studies were included for analysis. None of the studies evaluated the effectiveness of primary PET screening specific to lung cancer. Eight studies focused on primary PET screening for all types of cancer; the detection rates of lung cancer were low. Four studies reported evidence of lung cancer screening programs with selective PET, in which the estimated pooled sensitivity and specificity was 83 % and 91 %, respectively. The authors concluded that the role of primary PET screening for lung cancer remains unknown. However, PET has high sensitivity and specificity as a selective screening modality. Moreover, they stated that further studies must be conducted to evaluate the use of PET or PET/CT screening for high-risk populations, preferably using randomized trials or prospective registration.

In a Cochrane review, Manser and colleagues (2013) examined if screening for lung cancer, using regular sputum examinations, CXR or CT scanning of the chest, reduces lung cancer mortality. These investigators searched electronic databases: the Cochrane Central Register of Controlled Trials (CENTRAL) (The Cochrane Library 2012, Issue 5), MEDLINE (1966 to 2012), PREMEDLINE and EMBASE (to 2012) and bibliographies. They also hand-searched the journal Lung Cancer (to 2000) and contacted experts in the field to identify published and unpublished trials. Controlled trials of screening for lung cancer using sputum examinations, CXR or chest CT were included in this analysis. These researchers performed an intention-to-screen analysis. Where there was significant statistical heterogeneity, they reported risk ratios (RRs) using the random-effects model. For other outcomes they used the fixed-effect model. These investigators included 9 trials in the review (8 RCTs and 1 controlled trial) with a total of 453,965 subjects. In one large study that included both smokers and non-smokers comparing annual CXR screening with usual care there was no reduction in lung cancer mortality (RR 0.99, 95 % CI: 0.91 to 1.07). In a meta-analysis of studies comparing different frequencies of CXR screening, frequent screening with CXR was associated with an 11 % relative increase in mortality from lung cancer compared with less frequent screening (RR 1.11, 95 % CI: 1.00 to 1.23); however several of the trials included in this meta-analysis had potential methodological weaknesses. These researchers observed a non-statistically significant trend to reduced mortality from lung cancer when screening with CXR and sputum cytology was compared with CXR alone (RR 0.88, 95 % CI: 0.74 to 1.03). There was one large methodologically rigorous trial in high-risk smokers and ex-smokers (those aged 55 to 74 years with greater than or equal to 30 pack-years of smoking and who quit less than or equal to 15 years prior to entry if ex-smokers) comparing annual LDCT screening with annual CXR screening; in this study the relative risk of death from lung cancer was significantly reduced in the LD CT group (RR 0.80, 95 % CI: 0.70 to 0.92). The authors concluded that the current evidence does not support screening for lung cancer with CXR or sputum cytology. Annual LD CT screening is associated with a reduction in lung cancer mortality in high-risk smokers; but further data are needed on the cost-effectiveness of screening and the relative harms and benefits of screening across a range of different risk groups and settings. This review does not mention the use of PET as a screening tool.

Furthermore, the American Cancer Society's guidelines on "Lung cancer screening" (Wender et al, 2013), the American College of Chest Physicians' clinical practice guidelines on "Screening for lung cancer: Diagnosis and management of lung cancer" (Detterbeck et al, 2013), as well as the National Comprehensive Cancer Network's clinical practice guideline on "Non-small cell lung cancer" (Version 3.2014) do not mention the use of PET as a screening tool.

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Nair and colleagues (2016) stated that PET is a diagnostic tool for lung cancer evaluation. No studies have ascertained practice patterns and determined the appropriateness of PET imaging in a large group of U.S. patients with screen-detected lung nodules. These investigators analyzed participants in the National Lung Screening Trial (NLST) with positive screening test results and identified individuals with a PET scan performed prior to lung cancer diagnosis (diagnostic PET). Appropriate scan was defined as one performed in a patient with a nodule greater than or equal to 0.8 cm. Logistic regression was used to assess factors associated with diagnostic PET scan use and appropriateness of PET scan use. Diagnostic PET imaging was performed in 1,556 of 14,195 patients (11 %) with positive screen results; 331 of these (21 %) were inappropriate; PET scan use by endemic fungal disease area was comparable although patients from the Northeast/Southeast were twice as likely as the West to have a diagnostic PET. Trial arm, older age, sex, nodule size greater than or equal to 0.8 cm, upper lobe location, and spiculated margin were variables positively associated with use. Trial arm, older age, and spiculated margin were positively associated with appropriate use. Only 561 diagnostic PETs (36 %) were recommended by a radiologist and 284 PETs performed for nodules less than 0.8 cm (86 %) were ordered despite no recommendation from a radiologist. The authors concluded that PET imaging was differentially used in the NLST and inappropriately used in many cases against radiologist recommendations. They stated that these data suggested PET imaging may be overused in the lung cancer screening population and may contribute to excess health-care costs.

Low-Dose Computed Tomography as a Screening Test for Asbestos-Exposed Individuals

Murray and colleagues (2016) noted that CT-based studies of asbestos-exposed individuals reported a high prevalence of lung cancer, but the utility of LDCT to screen asbestos-exposed populations is not established. These researchers discussed the prevalence of indeterminate pulmonary nodules and incidental findings on chest LDCT of asbestos-exposed individuals in Western Australia. A total of 906 subjects from the Western Australian Asbestos Review Program underwent LDCT of the chest as part of regular annual review. An indeterminate (solid) nodule was defined as greater than 50 mm3 and part-solid/non-solid nodules greater than 5 mm3. The presence of asbestos-related diseases was recorded with a standardized report; 58 (6.5 %) participants were current smokers, 511 (56.4 %) ex-smokers, and 325 (36.4 %) never-smokers; 104 indeterminate nodules were detected in 77 subjects (8.5 %); of these, 8 cases had confirmed lung cancer (0.88 %); 87 subjects (9.6 %) had incidental findings that required further investigation, 42 (4.6 %) from lower airways inflammation. The majority of nodules were solid, 4 to 6 mm3 and more common with age; 580 (64 %) subjects had pleural plaques, and 364 (40.2 %) had evidence of interstitial lung disease. The authors concluded that the prevalence of LDCT-detected indeterminate lung nodules in individuals with significant asbestos exposure from Western Australian was 8.5 %, with clinically important incidental findings in 9.4 %, predominantly related to lower respiratory (tract) inflammation. They stated that LDCT appeared to effectively describe asbestos-related diseases and is likely to be an acceptable modality to monitor asbestos-exposed individuals although there is a need to better define the level of risk of lung cancer before widespread adoption of such practice.

This study had several drawbacks. First, 80 % of the subjects having been pre-screened with chest X-ray at least 1 year prior to the LDCT, possibly creating a selection bias. Secondly, as this study was part of a clinical service, the initial scans were read by a 1 of 2 thoracic radiologists. However, any case with an indeterminate nodule, or any other feature of concern, was discussed at a multi-disciplinary team meeting (with further expert radiology and respiratory specialists) for consensus decision on further management. Finally, the validity of ultra-LDCT as a substitute for standard dose or high resolution CT of the thorax for the diagnosis of most lung, pleural and incidental abnormalities related to asbestos exposure has not been established.

The U.S. Preventive Services Task Force's webpage on "Lung cancer: Screening" (USPSTF, 2013) did not make a recommendation regarding the use of LDCT for asbestos exposure.

The Centers for Disease Control and Prevention's webpage on "Lung cancer" (CDC, 2018) does not list asbestos exposure as an indication for LDCT.

Furthermore, an UpToDate review on "Asbestos-related pleuropulmonary disease" (King, 2018) states that "High resolution computed tomography (HRCT) is more sensitive than plain films in detecting parenchymal

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abnormalities in asbestos-exposed individuals. Up to 30 % of asbestos-exposed individuals demonstrate an abnormal HRCT in spite of a normal chest radiograph. However, HRCT may still appear normal or near normal in cases of histopathologically proven asbestosis … The significance of abnormal HRCT scans in asymptomatic asbestos-exposed persons as well as the proper role of HRCT in detection of asbestos-induced lung disease need further study".

Low-Dose Computed Tomography for Prediction of Cardiovascular Event in Heavy Smokers

Garg and colleagues (2018) stated that evaluation of coronary artery calcification (CAC) during lung cancer screening chest CT represents an opportunity to identify asymptomatic individuals at increased coronary heart disease (CHD) risk. These researchers determined the improvement in CHD risk prediction associated with the addition of CAC testing in a population recommended for lung cancer screening. They included 484 out of 6,814 Multi-Ethnic Study of Atherosclerosis (MESA) subjects without baseline cardiovascular disease who met USPSTF CT lung cancer screening criteria and underwent gated CAC testing; 10 year-predicted CHD risks with and without CAC were calculated using a validated MESA-based risk model and categorized into low (less than 5 %), intermediate (5 % to 10 %), and high (greater than or equal to 10 %). The net re-classification improvement (NRI) and change in Harrell's C-statistic by adding CAC to the risk model were subsequently determined. Of 484 included subjects (mean age of 65 years; 39 % women; 32 % black), 72 (15 %) experienced CHD events over the course of follow-up (median of 12.5 years). Adding CAC to the MESA CHD risk model resulted in 17 % more subjects classified into the highest or lowest risk categories and a NRI of 0.26 ($p=0.001$). The C-statistic improved from 0.538 to 0.611 ($p=0.01$). The authors concluded that CHD event rates were high in this lung cancer screening eligible population. These individuals represented a high-risk population who merit consideration for CHD prevention measures regardless of CAC score. These researchers stated that although overall discrimination remained poor with inclusion of CAC scores, determining whether those re-classified to an even higher risk would benefit from more aggressive preventive measures may be important.

Fan and Fan (2018) noted that CAC is a well-established predictor of cardio-vascular events (CVEs). These researchers examined if lung cancer screening CT-based CAC score has a good cost-effectiveness for predicting CVEs in heavy smokers. They carried out a literature search according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. PubMed, Embase, and Cochrane library databases were systematically searched for relevant studies that examined the association between lung cancer screening CT-based CAC and CVEs up to December 31, 2017. These investigators selected fixed-effect model for analysis of data heterogeneity. Statistical analyses were performed by using Review Manager Version 5.3 for Windows. A total of 4 RCTs with 5,504 subjects were included. These findings demonstrated that CVEs were significantly associated with the presence of CAC (RR 2.85, 95 % CI: 2.02 to 4.02, $p<0.00001$). Moreover, higher CAC score (defined as CAC score greater than 400 or greater than 1,000) was associated with a significant increased CVE count (RR 3.47, 95 % CI: 2.65 to 4.53, $p<0.00001$). However, the prevalence of CVEs was not different between male and female groups (RR 2.46, 95 % CI: 0.44 to 13.66, $p=0.30$). The authors concluded that CAC Agatston score evaluated by lung cancer screening CT had potential in predicting the likelihood of CVEs in the early stage without sexual difference. Thus, it may guide clinicians to intervene those heavy smokers with increased risk of CVEs earlier by CAC score through lung cancer screening CT. These investigators stated that well-designed, large studies are still needed.

The authors stated that this study had several drawbacks. First, their extracted data were not the original data. Although these researchers analyzed the studies by CAC classifications, it was impossible to adjust potential confounders including inflammatory factors. Moreover, there were insufficient data on body mass index (BMI) and pack-years of smoking for a reliable analysis in this study. Second, it is noteworthy that the majority of subjects recruited in this study were heavy smoking. It should be cautious to generalize the findings to non-smokers. Third, only 4 RCTs with 5,504 subjects were included in this analysis and none was double-blind.

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Lin and associates (2018) stated that cardiovascular risk assessment employs traditional risk factors to identify individuals who may benefit from primary prevention therapies. Incorporating non-traditional risk factors may improve traditional multi-variate risk assessment. On behalf of the USPSTF, these investigators evaluated evidence for the use of non-traditional risk factors – ankle-brachial index (ABI), high-sensitivity C-reactive protein (hsCRP), and CAC – in asymptomatic adults without known cardiovascular disease (CVD); 5 key questions address: clinical impact of non-traditional risk factor assessment versus traditional risk factor assessment with Framingham Risk Score (FRS) or Pooled Cohort Equations (PCE) (KQ1), performance of non-traditional risk factor assessment added to the FRS or PCE (KQ2), harms of non-traditional risk factor assessment (KQ3), and benefits (KQ4) and harms (KQ5) of non-traditional risk factor-guided therapy. The USPSTF will use this review to update prior recommendations on the use of non-traditional risk factors and the use of CVD risk assessment with the ABI. Medline, PubMed, and Cochrane Collaboration Registry of Controlled Trials through May 22, 2017, were employed to update existing systematic reviews supporting the previous USPSTF recommendations. These researchers screened 22,707 abstracts and 483 full-text articles against a priori inclusion criteria. For KQ1 and KQ4, they limited studies to trials reporting patient health outcomes. For KQ2, they included risk prediction studies comparing a base model with traditional risk factors (the FRS or PCE) to extended models also including 1 of the 3 non-traditional risk factors (ABI, hsCRP, and CAC) predicting CHD or CVD outcomes. For KQ3 and KQ5, they broadly included any study design examining harms of non-traditional risk assessment or non-traditional risk factor-guided therapy. All KQs were limited to studies of asymptomatic populations that were conducted in developed nations and published in the English language. Two investigators independently and critically appraised each article that met inclusion criteria using USPSTF's design-specific criteria, supplemented by the Checklist for Critical Appraisal and Data Extraction for Systematic Review of Prediction Modelling Studies (CHARMS) for risk prediction studies. Poor-quality studies were excluded. Data from fair- and good-quality trials were abstracted into standardized evidence tools in DistillerSR, with all data double-checked by a 2nd reviewer for accuracy. Due to the limited number of included studies and/or clinical heterogeneity of included studies, these investigators did not conduct meta-analyses; they graded the strength of the overall body of evidence for each KQ. For KQ1 and KQ4, outcomes included fatal and non-fatal CVD events (e.g., myocardial infarction [MI], cerebrovascular accident [CVA]) and all-cause mortality. For KQ2, outcomes included any measure of calibration (e.g., calibration plot, Hosmer-Lemeshow test) or overall performance (e.g., likelihood ratio tests, R2), discrimination (e.g., c-statistic/area under the curve [AUC]), or re-classification (e.g., net re-classification index [NRI]). For KQ3, outcomes comprised any harms, including radiation exposure due to CT imaging for CAC and down-stream health care utilization. For KQ5, outcomes included any serious adverse event (AE) as defined by the included study. These researchers included a total of 43 unique studies reported in 54 publications (some studies were included for multiple KQs): 1 study for KQ1, 33 studies for KQ2, 8 studies for KQ3, 4 studies for KQ4, and 3 studies for KQ5. Based on a smaller body of evidence, CAC consistently appeared to improve discrimination and re-classification in both published coefficient and model development studies; NRIs ranged from 0.084 to 0.35. The authors concluded that there was no direct evidence from adequately powered clinical impact trials comparing traditional cardiovascular risk assessment to risk assessment using non-traditional risk factors on patient health outcomes. The best available indirect evidence was mainly limited to studies evaluating the incremental value on discrimination and risk reclassification when adding ABI, hsCRP, or CAC to the FRS. They had much less evidence on the addition of these non-traditional risk factors to the PCE (compared to the FRS) and much less evidence to inform how these non-traditional risk factors improve calibration of traditional cardiovascular risk assessment. Thus, the value of non-traditional risk factors to correct the over- or under-prediction of traditional risk assessment went unanswered. Overall, ABI may improve discrimination and re-classification in women when the base model performed poorly. While CAC appeared to be the most promising non-traditional risk factor to improve discrimination and re-classification, it was based on a smaller body of evidence that lacked individual patient or participant data (IPD) meta-analyses; CAC may also result in additional down-stream testing/procedures, and it is unclear whether these sequelae represent a net benefit or harm to individuals. One large RCT showed that high-intensity statin therapy in individuals with elevated hsCRP and normal lipid levels could reduce CVD morbidity and mortality, but it was unclear if these benefits would not also be applicable to individuals with normal hsCRP. Furthermore, treatment guided by non-traditional risk factors has not been evaluated against treatment guided by traditional multi-variate cardiovascular risk assessment. The authors stated that well-designed prospective studies that are reflective of real-world practice are needed to evaluate the down-stream effects of CAC on cardiac imaging and re-vascularization, as well as incidental findings, since these are common. These

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include studies that aid in determining whether the identification of incidental findings, and/or increased health care utilization, is a net benefit or net harm.

The authors stated that this review had numerous drawbacks. First, these investigators focused on the 3 most promising non-traditional risk factors: ABI, hsCRP, and CAC. They also restricted their inclusion to English language studies and studies in developed countries, although they did not believe this restriction biased their review findings. Given the large volume of studies included for KQ2, these researchers made some explicit exclusions so as to focus on the most clinically relevant analyses, such as the exclusion of: CVA-specific outcomes, CAC derived from lung cancer screening, or CT angiography, studies in which the comparator was a single non-traditional risk factor alone, and analyses that did not allow them to isolate the contribution of individual non-traditional risk factors (i.e., studies using base models including other risk factors and studies comparing the FRS to the RRS). Additionally, studies were excluded if it could not be determined whether re-classification was appropriate (i.e., re-classification was reported without respect to events). Additionally, the predictive value of traditional risk factors such as total or HDL cholesterol was taken as given, but some literature suggested that these, too, might be very small to small when assessed in terms of the c-statistic. These investigators were conservative in their data synthesis across the body of evidence; namely, they did not quantitatively pool c-statistics/AUC or NRI and they did not make direct comparisons of finding across studies. Even though the authors stratified their discussion by base model (the FRS versus PCE) and model type (published coefficients versus model development), many of the studies had variations in included populations (e.g., inclusion of patients with diabetes, distribution of CVD risk), differences in analyses (e.g., model re-calibration, time horizon), differences in outcomes predicted (e.g., hard versus soft events), and definitions of risk strata that prohibited more definitive conclusions. These researchers did, however, examine differences in non-traditional risk factor performance in those studies that examined more than 1 non-traditional risk factor.

Artificial Intelligence-Based Imaging for Lung Cancer Screening

Baldwin et al (2020) stated that estimation of the risk of malignancy in pulmonary nodules detected by CT is central in clinical management. The use of artificial intelligence (AI) offers an opportunity to improve risk prediction. These researchers compared the performance of an AI algorithm, the lung cancer prediction convolutional neural network (LCP-CNN), with that of the Brock University model, recommended in U.K. guidelines. A data-set of incidentally detected pulmonary nodules measuring 5 to 15 mm was collected retrospectively from 3 U.K. hospitals for use in a validation study. Ground truth diagnosis for each nodule was based on histology (required for any cancer), resolution, stability or (for pulmonary lymph nodes only) expert opinion. There were 1,397 nodules in 1,187 patients, of which 234 nodules in 229 (19.3 %) patients were cancer. Model discrimination and performance statistics at pre-defined score thresholds were compared between the Brock model and the LCP-CNN. The AUC for LCP-CNN was 89.6 % (95 % CI: 87.6 to 91.5), compared with 86.8 % (95 % CI: 84.3 to 89.1) for the Brock model ($p \leq 0.005$). Using the LCP-CNN, these investigators found that 24.5 % of nodules scored below the lowest cancer nodule score, compared with 10.9 % using the Brock score. Using the pre-defined thresholds, these investigators found that the LCP-CNN gave 1 false negative (0.4 % of cancers), whereas the Brock model gave 6 (2.5 %), while specificity statistics were similar between the 2 models. The authors concluded that the LCP-CNN score exhibited better discrimination and allowed a larger proportion of benign nodules to be identified without missing cancers than the Brock model. This has the potential to substantially reduce the proportion of surveillance CT scans needed; thereby, saving significant resources.

The authors stated that by means of machine learning (ML), they have developed and externally validated a new risk prediction model that compared favorably with the most accurate multi-variable model in current usage and may also have a role to play in identifying low-risk nodules that do not need further surveillance. It is important that this model is further refined to include further attention to scan quality and incorporation of additional clinical and CT data. These researchers stated that this model is being tested in a prospectively collected cohort in routine clinical practice to examine if the results observed in this study are reproducible, and what the impact of better risk prediction has on the efficiency of nodule management, timely diagnosis. and quality of life (QOL) for patients.

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Chetan et al (2022) noted that predictions of the LCP-CNN AI model are analogous to the Brock model. A total of 10,485 lung nodules in 4,660 participants from the National Lung Screening Trial (NLST) were analyzed. Both manual and automated nodule measurements were inputted into the Brock model, and this was compared to LCP-CNN. The performance of an experimental AI model was tested following ablating imaging features in a manner analogous to removing predictors from the Brock model. First, the nodule was ablated leaving lung parenchyma only. Second, a sphere of the same size as the nodule was implanted in the parenchyma. Third, internal texture of both nodule and parenchyma was ablated. Automated axial diameter (AUC 0.883) and automated equivalent spherical diameter (AUC 0.896) significantly improved the accuracy of Brock model when compared to manual measurement (AUC 0.873), although not to the level of the LCP-CNN (AUC 0.936). Ablating nodule and parenchyma texture (AUC 0.915) resulted in a small decrease in AI predictive accuracy, as did implanting a sphere of the same size as the nodule (AUC 0.889). Ablating the nodule leaving parenchyma only resulted in a large decrease in AI performance (AUC 0.717). The authors concluded that feature ablation was a feasible technique for understanding AI model predictions. Nodule size and morphology play the largest role in AI prediction, with nodule internal texture and background parenchyma playing a limited role. This is broadly analogous to the relative importance of morphological factors over clinical factors within the Brock model. These researchers stated that these findings have important implications for future work on understanding AI prediction.

The authors stated that this study had several drawbacks. First, the accuracy of the Brock model in this study was somewhat lower than in the PanCan and BCCA cohorts used to develop Brock (AUC > 0.90). Similar differences have been reported in another secondary analysis of NLST data using the Brock model and were likely due to underlying differences between cohorts. Second, the predictive value of the Brock model is contingent on the prevalence of lung cancer in the population. This may differ in a clinical cohort from the 5.5 % to 5.6 % observed in screening cohorts such as NLST, PanCan, and BCCA. Third, the selection criteria applied to the NLST cohort and to this analysis limited the generalizability to clinical practice. Patients outside the age of 55 to 75 years, with previous lung cancer, recent chest CT, hemoptysis, or unexplained weight loss were all excluded from NLST. Nodules measuring less than 6 mm and ground glass opacities were excluded from this analysis. Patients with incidental pulmonary nodules in clinical practice could fall outside these criteria. Fourth, the manual diameter measurements from NLST were used directly, rather than being performed again, which may have resulted in bias in the predictive performance for the manual measurement. Fifth, there was likely bias when comparing LCP-CNN and Brock in this analysis as LCP-CNN was trained using data from NLST while Brock was trained on a separate population. However, direct comparison of the 2 models was not the main objective of this study; and would require testing in a previously unseen population. Finally, ablated CT images are atypical images that are not seen in clinical practice and that are challenging to interpret. It was difficult to attribute the effects of ablation to 1 single factor, e.g., translating by 15 mm in order to ablate a nodule may, in part, reduce predictive accuracy because the local severity of emphysema is altered. These researchers stated that future investigation performing feature removal using different techniques is needed to draw stronger conclusions.

de Margerie-Mellon and Chassagnon (2023) stated that AI is a broad concept that usually refers to computer programs that can learn from data and perform certain specific tasks. In recent years, the growth of deep learning (DL), a successful technique for computer vision tasks that does not require explicit programming, coupled with the availability of large imaging databases fostered the development of multiple applications in the medical imaging field, especially for lung nodules and lung cancer, mostly via CNN. Some of the 1st applications of AI in this field were dedicated to automated detection of lung nodules on X-ray and CT examinations, with performances now reaching or exceeding those of radiologists. For lung nodule segmentation, CNN-based algorithms applied to CT images showed excellent spatial overlap index with manual segmentation, even for irregular and ground glass nodules. A 3rd application of AI is the classification of lung nodules between malignant and benign, which could limit the number of follow-up CT examinations for less suspicious lesions. Several algorithms have reported excellent capabilities for the prediction of the malignancy risk when a nodule is discovered. These different applications of AI for lung nodules are especially appealing in the context of lung cancer screening. In the field of lung cancer, AI tools applied to lung imaging have been examined for distinct aims. First, they could play a role for the non-invasive characterization of tumors, especially for histological subtype and somatic mutation predictions, with a potential therapeutic impact. Furthermore, they could aid in predicting the patient prognosis, in combination to clinical data. Moreover, these investigators stated that despite these encouraging

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perspectives, clinical implementation of AI tools is only beginning because of the lack of generalizability of published studies, of an inner obscure working and because of limited data regarding the impact of such tools on the radiologists' decision and on the patient outcome. The authors concluded that radiologists must be active participants in the process of evaluating AI tools, as such tools could support their daily work and offer them more time for high added value tasks.

Landy et al (2023) noted that annual low-dose CT (LDCT) screening reduces lung cancer mortality; however, harms could be reduced, and cost-effectiveness improved by re-using the LDCT image in conjunction with DL or statistical models to identify low-risk individuals for biennial screening. These researchers identified low-risk individuals in the NLST; and estimated had they been assigned a biennial screening; how many lung cancers would have been delayed 1 year in diagnosis. This diagnostic study included participants with a presumed non-malignant lung nodule in the NLST between January 1, 2002, and December 31, 2004, with follow-up completed on December 31, 2009. Data were analyzed for this study from September 11, 2019, to March 15, 2022. An externally validated DL algorithm that predicts malignancy in current lung nodules using LDCT images (LCP-CNN; Optellum Ltd) was re-calibrated to predict 1-year lung cancer detection by LDCT for presumed non-malignant nodules. Individuals with presumed non-malignant lung nodules were hypothetically assigned annual versus biennial screening based on the re-calibrated LCP-CNN model, Lung Cancer Risk Assessment Tool (LCRAT + CT [a statistical model combining individual risk factors and LDCT image features]), and the American College of Radiology (ACR) recommendations for lung nodules, version 1.1 (Lung-RADS). Primary outcomes included model prediction performance, the absolute risk of a 1-year delay in cancer diagnosis, and the proportion of individuals without lung cancer assigned a biennial screening interval versus the proportion of cancer diagnoses delayed. The study included 10,831 LDCT images from patients with presumed non-malignant lung nodules (58.7 % men; mean [SD] age of 61.9 [5.0] years), of whom 195 were diagnosed with lung cancer from the subsequent screen. The re-calibrated LCP-CNN had substantially higher AUC (0.87) than LCRAT + CT (0.79) or Lung-RADS (0.69) to predict 1-year lung cancer risk ($p < 0.001$). If 66 % of screens with nodules were assigned to biennial screening, the absolute risk of a 1-year delay in cancer diagnosis would have been lower for re-calibrated LCP-CNN (0.28 %) than LCRAT + CT (0.60 %; $p = 0.001$) or Lung-RADS (0.97 %; $p < 0.001$). To delay only 10 % of cancer diagnoses at 1 year, more people would have been safely assigned biennial screening under LCP-CNN than LCRAT + CT (66.4 % versus 40.3 %; $p < 0.001$). The authors concluded that in this diagnostic study evaluating models of lung cancer risk, a re-calibrated DL algorithm was most predictive of 1-year lung cancer risk and had least risk of 1-year delay in cancer diagnosis among individuals assigned biennial screening. These researchers stated that DL algorithms could prioritize individuals for work-up of suspicious nodules and decrease screening intensity for individuals with low-risk nodules, which may be vital for implementation in healthcare systems.

The authors stated that this study had several drawbacks. Although 2 randomized clinical trials suggested that biennial screening was safe, the reduction in screening effectiveness from biennial screening is unknown. Within the NLST, data did not exist to identify whether a cancer developed from any specific nodule; thus, it was not possible to definitively know specifically which (or whether any) of the nodules evaluated by LCP-CNN progressed to cancer. The LCP-CNN model required the nodule to be identified before it can be applied, and these investigators only examined its performance in solid nodules. Currently, the NLST is the only large-scale lung screening study whose images are publicly available, and external validation is important before this approach could be implemented clinically. Although LCP-CNN was developed using some NLST images, which could potentially bias the results toward over-estimating its performance, the LCP-CNN has performed well in external validation studies; these researchers used only the cross-validated LCP-CNN predictor that left out the image in question, although other images from the same individual could be included; and these researchers used an optimism-adjusted AUC. The LCRAT + CT model was also developed using NLST data; similarly, these investigators used cross-validation and optimism-adjusted AUCs to reduce the potential bias. Although LCP-CNN has been externally validated, the single re-calibration parameter for re-calibrated LCP-CNN might need to be re-calculated for use in populations with substantially different lung cancer risk. The Lung-RADS score was updated in November 2022; while most nodules would be assigned the same Lung-RADS score, the authors did not have the raw data needed to re-score each nodule. Other drawbacks reduced the predictive abilities of the models. These researchers had no LCP-CNN score for screens containing only 4-mm nodules (as measured by Optellum Ltd), which have the least risk and are the best candidates for interval extension; the performance of LCP-

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CNN would likely improve if all nodule sizes were evaluated. These investigators did not have nodule volume to input into LCRAT + CT, which may further improve its performance. Improvements in imaging technology since the NLST meant that the scans examined in this study did not represent current practice, and may also improve the performance of LCRAT + CT and LCP-CNN.

Paez et al (2023) stated that a DL model (LCP CNN) for the stratification of indeterminate pulmonary nodules (IPNs) showed better discrimination than commonly used clinical prediction models; however, the LCP CNN score was based on a single time-point that ignores longitudinal information when prior imaging studies are available. Clinically, IPNs are often followed over time and temporal trends in nodule size or morphology inform management. These investigators examined if the change in LCP CNN scores over time was different between benign and malignant nodules. This study used a prospective-specimen collection, retrospective-blinded-evaluation (PRoBE) design. Subjects with incidentally or screening detected IPNs of 6 to 30 mm in diameter with at least 3 consecutive CT scans before diagnosis (slice thickness 1.5 mm or less) with the same nodule present were included. Disease outcome was adjudicated by biopsy-proven malignancy, biopsy-proven benign disease, and absence of growth on at least 2-year imaging follow-up. Lung nodules were analyzed using the Optellum LCP CNN model. Investigators performing image analysis were blinded to all clinical data. The LCP CNN score was determined for 48 benign and 32 malignant nodules. There was no significant difference in the initial LCP CNN score between benign and malignant nodules. Overall, the LCP CNN scores of benign nodules remained relatively stable over time while that of malignant nodules continued to increase over time. The difference in these 2 trends was statistically significant. These researchers also developed a joint model that incorporates longitudinal LCP CNN scores to predict future probability of cancer. Malignant and benign nodules appeared to have distinctive trends in LCP CNN score over time. The authors concluded that the findings of this study suggested that longitudinal modeling may improve radiomic prediction of lung cancer over current models; further studies are needed to validate these early findings.

The authors stated that this study had several drawbacks including the convenient sample used, the limited sample size, and the lack of a validation dataset. The study population was composed of patients with multiple CT scans available for analysis within the research biorepository; thus, was not necessarily representative of a particular clinical population. Furthermore, as routinely happens in clinical practice, the intervals between scans were highly variable and many subjects did not have scans at 3 and 6 months. Therefore, the model discriminatory ability was affected at these early time-points. Similarly, many subjects did not have more than 2 years follow-up, and this could also affect the model's performance. These drawbacks could be overcome in future studies by enrolling subjects with shorter and longer interval follow-ups. Another potential drawback and source of error was the use of different CT scanners and acquisition protocols; however, these researchers only included non-contrast CT scans with slice thickness of1.5 mm or less. In addition, the LCP-CNN model was trained in the NLST data-set and validated in multiple cohorts from 2 different countries, which would represent numerous CT scanners and protocols. These investigators did not have the radiologist assessment or recommendation for many of the CT scans as they were from outside institutions and radiology reports were unavailable. It was possible that the radiologist assessment of the nodule could be as good or better than the LCP-CNN model, although a recent study showed that the LCP-CNN score improved readers' ability to correctly classify pulmonary nodules as benign or malignant6. Because the analysis was based on a limited retrospective dataset, these researchers could only infer from a separation of the probability score trajectories that occurred before the time of diagnosis that such longitudinal analysis would in reality result in earlier diagnosis, treatment, and improved patient outcomes. It was possible that these trends were introduced by biases inherent in the retrospective design of this trial since patients present to lung nodule experts at widely different time-points in the course of their disease. These investigators stated that future work based on prospective evaluation of these trends are needed to confirm their hypothesis.

Thong et al (2023) noted that lung cancer is the main cause of cancer-related deaths worldwide. Early detection of lung cancer with screening is crucial to lower the high morbidity and mortality rates. Artificial intelligence (AI) is widely used in healthcare, including in the assessment of medical images. A growing number of reviews studied the use of AI in lung cancer screening; however, no over-arching meta-analysis has examined the diagnostic test accuracy (DTA) of AI-based imaging for lung cancer screening. These investigators systematically reviewed the DTA of AI-based imaging for lung cancer screening. PubMed,

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Embase, Cochrane Library, CINAHL, IEEE Xplore, Web of Science, ACM Digital Library, Scopus, PsycINFO, and ProQuest Dissertations and Theses were searched from inception to present day. Studies that were published in English and that examined the performance of AI-based imaging for lung cancer screening were included. Two independent reviewers screened titles and abstracts and employed the Quality Assessment of Diagnostic Accuracy Studies-2 tool to evaluate the quality of selected studies. Grading of Recommendations Assessment, Development, and Evaluation (GRADE) to diagnostic tests was used to assess the certainty of evidence. A total of 26 studies with 150,721 imaging data were included. Hierarchical summary ROC model used for meta-analysis demonstrated that the pooled sensitivity for AI-based imaging for lung cancer screening was 94.6 % (95 % CI: 91.4 % to 96.7 %) and specificity was 93.6 % (95 % CI: 88.5 % to 96.6 %). Subgroup analyses revealed that similar results were found among different types of AI, region, data source, and year of publication, but the overall quality of evidence was very low. The authors concluded that AI-based imaging could effectively detect lung cancer and be incorporated into lung cancer screening programs. Moreover, these researchers stated that certainty of evidence was very low; and further high-quality DTA studies on large lung cancer screening populations are needed to validate AI's role in early lung cancer detection.

An UpToDate review on “Overview of the initial evaluation, diagnosis, and staging of patients with suspected lung cancer” (Thomas et al, 2024) does not mention artificial intelligence / machine learning as a management option.

Furthermore, National Comprehensive Cancer Network’s clinical practice guideline on “Lung Cancer Screening” (Version 12.2024) does not mention artificial intelligence / machine learning as a screening tool.

Low-Dose CT for Surveillance of Patients with Non-Small Cell Lung Cancer who have Undergone Treatment

Jaklitsch et al (2012) noted that lung cancer is the leading cause of cancer death in North America. Low-dose CT screening can reduce lung cancer-specific mortality by 20 %. The American Association for Thoracic Surgery (AATS) created a multi-specialty task force to create screening guidelines for groups at high-risk of developing lung cancer and survivors of previous lung cancer. The AATS guidelines called for annual lung cancer screening with low-dose CT screening for North Americans from age of 55 to 79 years with a 30 pack-year history of smoking. Long-term lung cancer survivors should have annual low-dose CT to detect 2nd primary lung cancer until the age of 79 years. Annual low-dose CT lung cancer screening should be offered starting at age 50 years with a 20 pack-year history if there is an additional cumulative risk of developing lung cancer of 5 % or greater over the following 5 years. Lung cancer screening requires participation by a subspecialty-qualified team. The AATS will continue engagement with other specialty societies to refine future screening guidelines. The authors concluded that the AATS provided specific guidelines for lung cancer screening in North America.

An UpToDate review on “Management of stage I and stage II non-small cell lung cancer” (Vallieres and Schild, 2024) states that “Post-Therapy Surveillance – The rationale for surveillance following the initial treatment of NSCLC is for early detection of recurrent disease or a second primary lung cancer. A history, physical examination, and chest computed tomography (CT) are suggested every 6 months during the first 2 years after treatment and annually thereafter, although there are no data from randomized trials supporting the value of CT. However, it should also be noted that many of these patients have a smoking history and would potentially benefit from screening CTs even if they had not been treated for lung cancer, based on age and number of pack-years”.

Furthermore, National Comprehensive Cancer Network’s clinical practice guideline on “Non-small cell lung cancer” (Version 2.2024) states that “The NCCN NSCLC Panel feels that low-dose CT is beneficial for identifying recurrences in patients previously treated for NSCLC. It is important to note that the surveillance recommendations for patients who have been treated for NSCLC are different from the screening recommendations for individuals at high risk for lung cancer. The NCCN Guidelines recommend a chest CT scan with (or without) contrast and an history and physical (H&P) for the initial surveillance schedules (2 to 5 years after definitive treatment) followed by annual low-dose non-contrast-enhanced CT and an H&P”.

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